Geometry Foundations

Shef Scholars Competitive Math Academy

Week 0

Introduction

Main idea/Fact (What Week 0 is for)

Week 0 is not new material. It is the floor every later lecture stands on.

Three groups of facts carry almost every geometry problem in the first fifteen questions of the AMC 10: the angle rules, the length rules (triangle inequality, Pythagoras, and the two special right triangles), and the area and perimeter formulas for the standard figures. This handout collects all three, proves each one so you can rebuild it if you ever forget it, and then drills it.

The goal is recognition speed. By November you should not be *recalling* that an exterior angle equals the sum of the two remote interior angles. You should be seeing it in the figure.

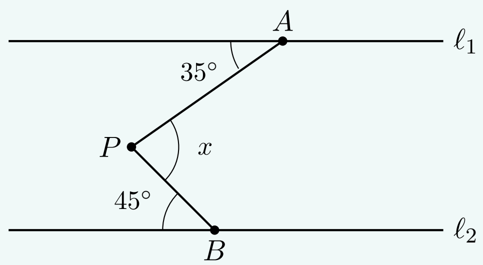
Work every problem before looking at a hint. Each problem has two hints: the first names the tool, the second moves you forward without handing over the answer.

A problem to start

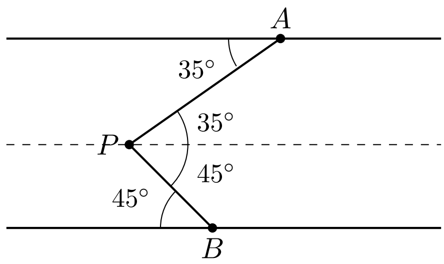
Geometry on the AMC 10 starts with angle chasing. Here is the kind of problem this handout will make routine.

Example 1

In the figure, lines ℓ_1 and ℓ_2 are parallel. Find the angle x.



Solution. The point P sits between the two parallel lines but does not lie on either of them, so no rule applies to it directly. We make a rule apply: draw a third line through P, parallel to both.



The dashed line and ℓ_1 are parallel, and the segment AP crosses both. So the $35°$ at A reappears at P, above the dashed line. In the same way, the dashed line and ℓ_2 are parallel, so the $45°$ at B reappears at P, below the dashed line. The angle x is exactly these two pieces put together:

$$x = 35° + 45° = \boxed{80°}.$$

Remark. Notice that we never computed anything clever. We added one line to the picture, and the picture solved itself. Auxiliary lines, especially auxiliary *parallel* lines, are the first real technique of contest geometry, and by the end of Part I every rule they rely on will be yours.

The main habit

Main idea/Fact (Main habit for geometry problems)

Draw a large figure and write every fact on it. Each time you find a new angle or length, mark it on the figure immediately, then ask: what does this newly unlock?

Angle chasing is not one big deduction. It is a chain of tiny ones, and the figure is where the chain lives. Students who keep the angles in their head lose track by the third step. Students who mark the figure can see the next step waiting.

Pitfall

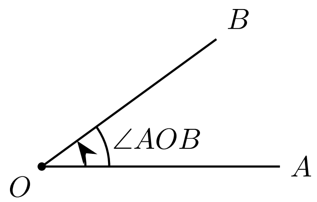
AMC figures are frequently *not drawn to scale*, and the test says so. Never measure a figure, and never conclude that two segments are equal or two lines are parallel because they look it. Use only what is given or what you can prove.

Points, Lines, and Angles

Points, lines, and the plane

In mathematics, a **plane** is a flat two-dimensional surface extending indefinitely in every direction. Your sheet of paper is a good model of a piece of one. A **point** is a location in the plane, and through any two distinct points passes exactly one **line**.

> **Definition** (Angle)
>
> An **angle** is the amount of *turn* from one ray to another ray sharing the same endpoint, called the **vertex**. We write $\angle AOB$ for the angle at vertex O between rays OA and OB: the vertex letter always sits in the middle.

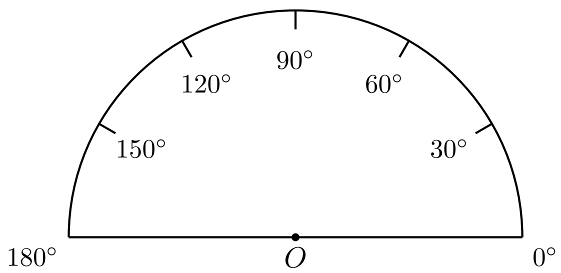


Question for you: if ray OB rotated further away from ray OA (counterclockwise), would the angle get bigger or smaller?

We measure angles in **degrees** (°). A full turn is $360°$, so a half turn, which is a straight line, is $180°$, and a quarter turn is $90°$.

A measuring picture

Think of $0°$ as no turn at all, and $180°$ as the half turn that lands you on a straight line.



> **Definition** (Kinds of angles)
>
> An angle is **acute** if it is less than $90°$, **right** if it is exactly $90°$, **obtuse** if it is between $90°$ and $180°$, and **straight** if it is exactly $180°$. In figures, a right angle is marked with a small square instead of an arc.
>
> Two lines meeting at a right angle are **perpendicular**, written $\ell_1 \perp \ell_2$.

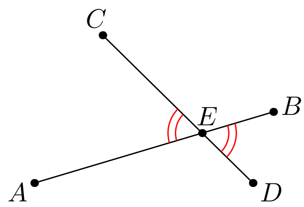
> **Definition** (Complementary and supplementary)
>
> Two angles are **complementary** if they sum to $90°$, and **supplementary** if they sum to $180°$. Both words appear in contest statements, so learn them as vocabulary.

Main idea/Fact (Angles on a line, angles at a point)
Angles that together fill one side of a straight line sum to $180°$. Angles that together surround a point sum to $360°$.

These two facts do more angle chasing than everything else combined.

Vertical angles

When two lines cross, four angles appear, and they come in equal pairs.



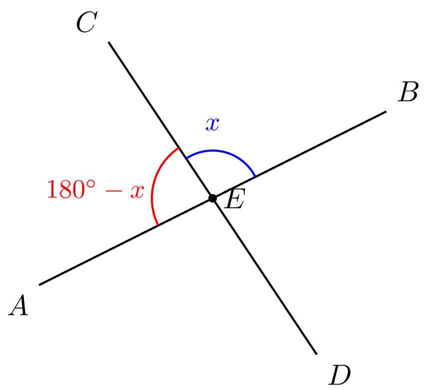
Main idea/Fact (Vertical angles)
When lines AB and CD intersect at E, the opposite pairs are equal:

$$\angle AEC = \angle BED \qquad \text{and} \qquad \angle AED = \angle BEC.$$

These pairs are called **vertical angles** (also called opposite angles).

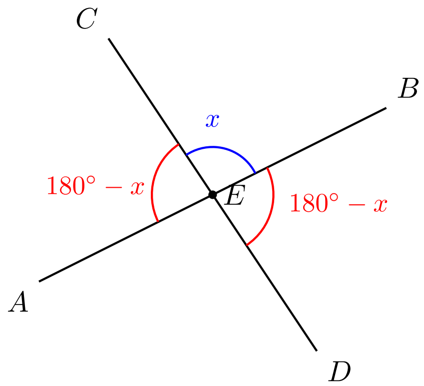
Why this is true. Let $\angle CEB = x$. Since A, E, B lie on one straight line,

$$\angle CEA + \angle CEB = 180°, \qquad \text{so} \qquad \angle CEA = 180° - x.$$

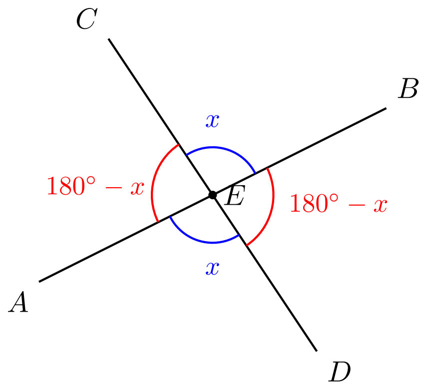


Now C, E, D also lie on one straight line, so

$$\angle CEB + \angle BED = 180°, \qquad \text{so} \qquad \angle BED = 180° - x.$$



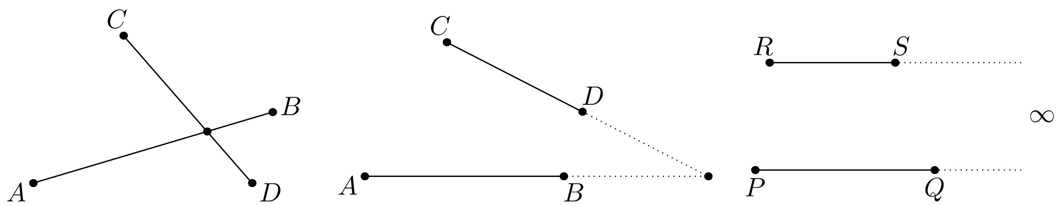
Therefore $\angle CEA = \angle BED = 180° - x$, and the same reasoning gives $\angle CEB = \angle AED = x$.



Remark. Notice the shape of that argument. We did not memorize four equalities, we used the straight-line fact twice. Almost everything in angle chasing decomposes into the 180° fact used repeatedly.

Parallel lines and transversals

> **Definition** (Parallel lines)
>
> Two lines in the plane are **parallel** if they never intersect, no matter how far they are extended. We write $PQ \parallel RS$. You can imagine parallel lines as lines that only "meet at infinity".



In the first two panels, AB and CD eventually intersect, so they are not parallel. In the third panel, PQ and RS never meet, so $PQ \parallel RS$.

A line that crosses two parallel lines is called a **transversal**, and the crossing creates equal angles everywhere.

Main idea/Fact (Parallel lines and a transversal)

Let $AB \parallel CD$, and let a transversal GH meet CD at E and AB at F. Then

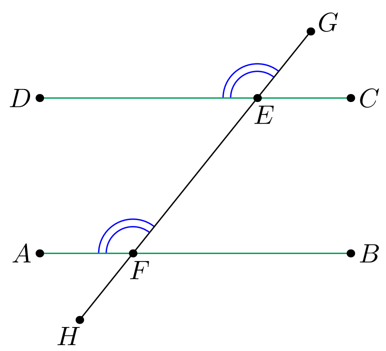
$$\angle DEG = \angle AFG.$$

Combining this with vertical angles and the straight-line fact, the eight angles in the figure form just two values:

$$\angle DEG = \angle AFG = \angle BFH = \angle CEH,$$
$$\angle CEG = \angle BFG = \angle AFH = \angle DEH,$$

and any angle from the first list plus any angle from the second sums to 180°.

The *converse* also holds: if a transversal makes equal angles of this kind with two lines, the two lines are parallel.



You may know these pairs by their school nicknames: equal angles in the same position at each crossing are **corresponding angles** (an F shape), equal angles on opposite sides between the lines are **alternate angles** (a Z shape), and same-side angles between the lines are **co-interior angles**, which sum to 180° (a C shape). The names do not matter on the contest. The picture does.

Main idea/Fact (The auxiliary parallel line)

If a point sits between two parallel lines but on neither, draw a third parallel line through it. Every angle at that point then splits into two pieces governed by the parallel rules, exactly as in the opening problem.

Pitfall

The parallel-line rules require the lines to actually be parallel, either given in the problem or proven by you. Two lines that merely look parallel in the figure earn you nothing.

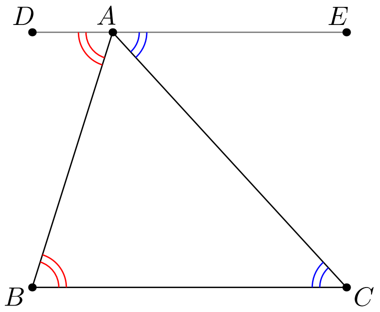
The angles of a triangle

Three points not on a common line form a **triangle**.

Main idea/Fact (Angle sum of a triangle)
The three interior angles of any triangle sum to $180°$:

$$\angle A + \angle B + \angle C = 180°.$$

Why this is true. Draw the line DE through A parallel to BC.



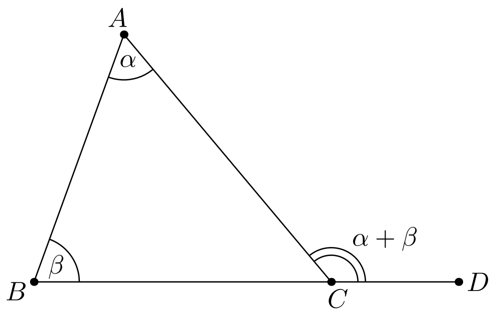
Since $DE \parallel BC$, the transversal AB gives $\angle ABC = \angle BAD$, and the transversal AC gives $\angle ACB = \angle CAE$. The three angles $\angle BAD$, $\angle BAC$, $\angle CAE$ sit together along the straight line DE, so

$$\angle A + \angle B + \angle C = \angle BAC + \angle BAD + \angle CAE = \angle DAE = 180°.$$

Notice this is the auxiliary parallel line again. One extra line turned a fact about a triangle into the straight-line fact.

Main idea/Fact (Exterior angle theorem)
Extend one side of a triangle. The **exterior angle** created equals the sum of the two remote interior angles:

$$\angle ACD = \alpha + \beta.$$



Why this is true. The interior angle at C is $180° - \alpha - \beta$ by the angle sum, and the exterior angle is $180°$ minus the interior one, which is $\alpha + \beta$.

This shortcut saves a step constantly: if the two remote interior angles are $47°$ and $62°$, the exterior angle is $109°$ immediately, with no need to find the third angle first.

Exercise 2. Two angles of a triangle are $35°$ and $80°$. Find the third angle, and find the exterior angle at the $80°$ vertex.

Solution. The third angle is $180° - 35° - 80° = 65°$. The exterior angle at the $80°$ vertex is $180° - 80° = 100°$, which indeed equals $35° + 65°$. ◇

Equal sides give equal angles

Definition (Kinds of triangles)

By sides: a triangle is **scalene** if all sides differ, **isosceles** if at least two sides are equal, and **equilateral** if all three are equal.

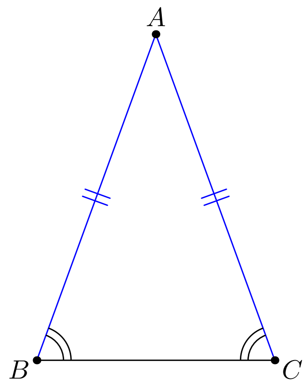
By angles: a triangle is **acute** if all angles are less than 90°, **right** if one angle is exactly 90°, and **obtuse** if one angle exceeds 90°.

Main idea/Fact (The isosceles triangle, in both directions)

In a triangle ABC,

$$AB = AC \iff \angle ACB = \angle ABC.$$

Equal sides force the angles opposite them to be equal, and equal angles force the sides opposite them to be equal. You will use the arrow in both directions constantly.



Main idea/Fact (The apex line does three jobs)

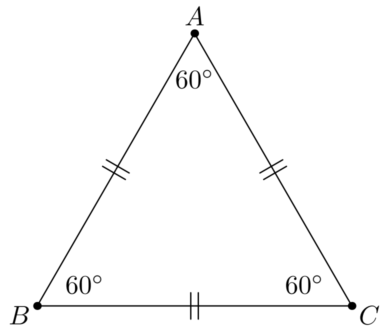
In an isosceles triangle ABC with $AB = AC$, the segment from A perpendicular to BC is simultaneously the height to BC, the median to BC (so it hits the midpoint), and the bisector of $\angle BAC$.

This one fact is why almost every isosceles computation starts with "drop the altitude from the apex".

Example 3

Prove that each angle of an equilateral triangle is 60°.

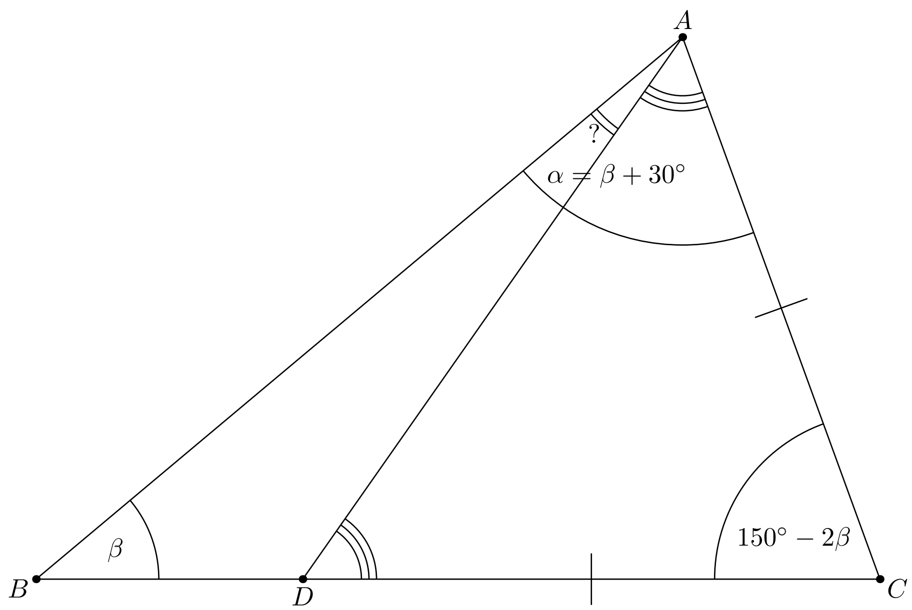
Solution. Apply the isosceles fact twice. From $AB = AC$ we get $\angle B = \angle C$, and from $AB = BC$ we get $\angle A = \angle C$. So all three angles are equal, and since they sum to 180°, each is 60°.



Here is a full angle chase that uses every tool so far.

Example 4

Let $\triangle ABC$ satisfy $\angle CAB - \angle ABC = 30°$. On side BC choose a point D with $AC = CD$. Find $\angle DAB$.



Solution. Put everything in terms of one variable. Let $\angle CAB = \alpha$ and $\angle ABC = \beta$, so $\alpha = \beta + 30°$. The angle sum in $\triangle ABC$ gives

$$\beta + \angle BCA + (\beta + 30°) = 180°, \qquad \text{so} \qquad \angle BCA = 150° - 2\beta.$$

Now look at $\triangle ACD$. Since $AC = CD$, it is isosceles with apex C, so $\angle DAC = \angle ADC$ and

$$\angle DAC = \frac{180° - (150° - 2\beta)}{2} = 15° + \beta.$$

Therefore

$$\angle DAB = \angle CAB - \angle DAC = (30° + \beta) - (15° + \beta) = \boxed{15°}.$$

Remark. The unknown β cancelled. That is the signature of this problem type: when a problem gives you a *difference* of angles rather than the angles themselves, name a variable, push it through, and expect it to disappear at the end.

Polygons and angle sums

Definition (Polygon)

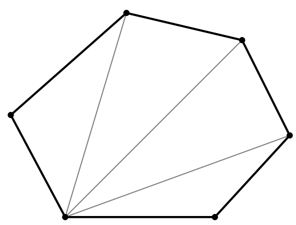
A **polygon** is a closed figure made of straight segments. A polygon with n vertices is an **n-gon**. A polygon is **convex** if all its interior angles are less than $180°$, and **regular** if all its sides are equal and all its angles are equal.

Main idea/Fact (Interior angle sum)

The interior angles of an n-gon sum to

$$(n-2)\cdot 180°.$$

Why this is true. Pick one vertex of a convex n-gon and draw the diagonals from it to every non-adjacent vertex. This cuts the polygon into $n-2$ triangles, and the interior angles of the polygon are exactly the angles of these triangles put together. Each triangle contributes $180°$.



For example, a hexagon has angle sum $(6-2)\cdot 180° = 720°$, so each angle of a *regular* hexagon is $720°/6 = 120°$.

Main idea/Fact (Exterior angle sum)

At each vertex of a convex polygon, extend one side to form an exterior angle. These exterior angles, one per vertex, always sum to

$$360°,$$

no matter how many sides the polygon has. In a regular n-gon, each exterior angle is therefore $\dfrac{360°}{n}$, and each interior angle is $180° - \dfrac{360°}{n}$.

Why this is true. At every vertex, interior plus exterior is $180°$. Summing over all n vertices gives $180°n$, and subtracting the interior sum $(n-2)\cdot 180°$ leaves $360°$. There is also a way to *see* it: walk once around the boundary of the polygon. At each corner you turn by the exterior angle, and one full trip around is one full turn.

Exercise 5. What is each interior angle of a regular octagon? Of a regular 12-gon?[1]

[1] Octagon: exterior $360°/8 = 45°$, so interior $135°$. Regular 12-gon: exterior $30°$, interior $150°$.

Pitfall

Almost every regular-polygon question is faster through the *exterior* angle. Going through the interior sum needs two steps and a division by n; going through the exterior angle needs one. Note also that $360°/n$ is the exterior angle only for *regular* polygons. For an irregular polygon, only the total of $360°$ is guaranteed.

Degrees, minutes, and seconds

Occasionally an angle is written in the older format

$$D^\circ\ M'\ S'' \qquad \text{where} \qquad 1^\circ = 60', \qquad 1' = 60''.$$

This is base 60, exactly like hours and minutes on a clock, so carrying and borrowing work the same way.

Example 6

Compute $37^\circ\,23'\,45'' + 21^\circ\,12'\,34''$ and $37^\circ\,23'\,45'' - 21^\circ\,12'\,54''$.

Solution. Adding column by column gives $58^\circ\,35'\,79''$. Since $79'' = 1'\,19''$, we carry one minute:

$$58^\circ\,36'\,19''.$$

For the subtraction we cannot take $54''$ from $45''$, so we borrow one minute: $23'$ becomes $22'$ and $45''$ becomes $105''$. Then

$$105'' - 54'' = 51'', \qquad 22' - 12' = 10', \qquad 37^\circ - 21^\circ = 16^\circ,$$

giving $\boxed{16^\circ\,10'\,51''}$.

Pitfall

The AMC states angles in plain degrees, so this notation is a ”know it if you see it” item rather than a drill target. Do not spend Week 0 practice time here.

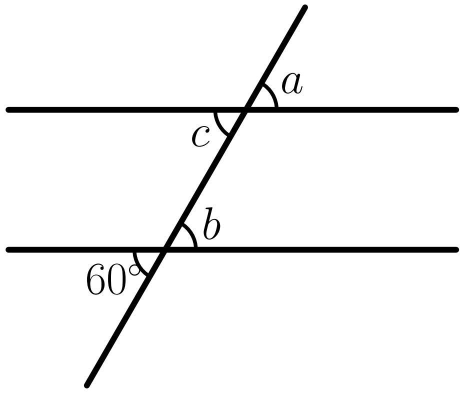
Summary of Part I

Main idea/Fact (Angle rules worth having on sight)

- Angles on a line sum to 180°; angles around a point sum to 360°; vertical angles are equal.
- Parallel lines plus a transversal produce only two angle values, and pairs from opposite families sum to 180°. Parallelism must be given or proven, never read off the picture.
- A point stuck between two parallel lines calls for an auxiliary parallel line through it.
- Triangle angles sum to 180°, and an exterior angle equals the sum of the two remote interior angles.
- $AB = AC \iff \angle B = \angle C$. In an isosceles triangle the apex perpendicular is also the median and the angle bisector.
- An n-gon's interior angles sum to $(n-2)\cdot 180^\circ$; its exterior angles sum to 360°; a regular n-gon has exterior angle $360^\circ/n$.

Problems: Points, Lines, and Angles

1. Determine the angles labeled with letters, given that the two horizontal lines are parallel.

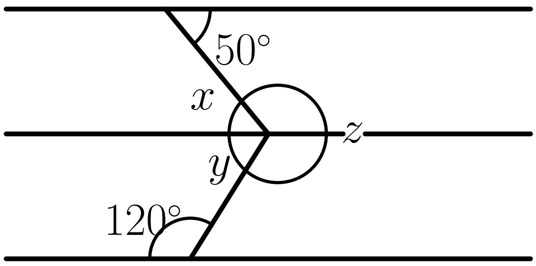


(Hint 1, Hint 2)

2. X is the midpoint of AB and Y is the midpoint of AX. If $BY = 9$, what is AB?

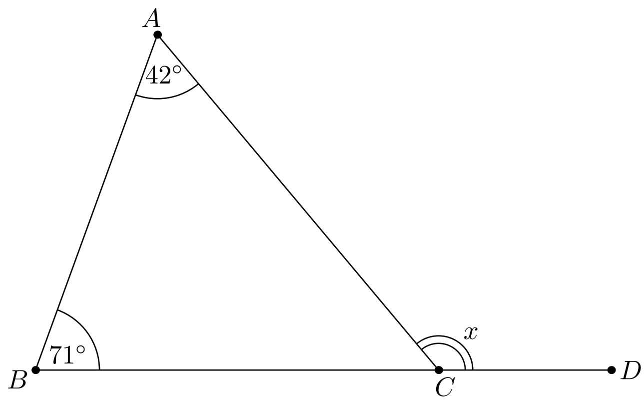
(Hint 1, Hint 2)

3. Determine the angles labeled with letters, given that the three horizontal lines are parallel.



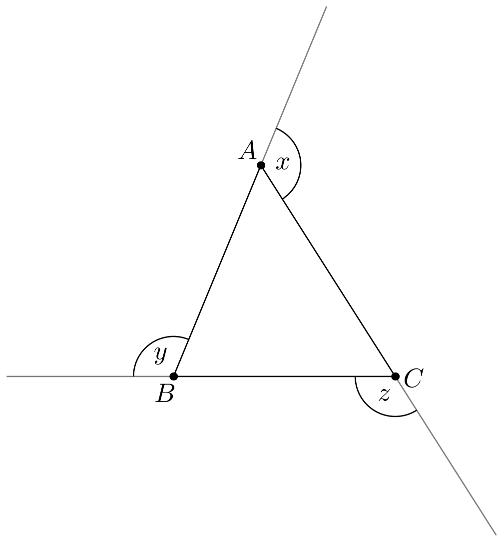
(Hint 1, Hint 2)

4. In the figure, B, C, D lie on a line in that order. Given $\angle BAC = 42°$ and $\angle ABC = 71°$, find x.



(Hint 1, Hint 2)

5. Each side of $\triangle ABC$ has been extended in one direction, as shown. Find $x + y + z$.



(Hint 1, Hint 2)

6. Each interior angle of a regular polygon measures $156°$. How many sides does it have?

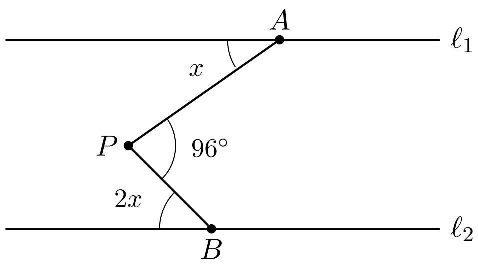
(Hint 1, Hint 2)

7. (2016 AMC 10B #7) The ratio of the measures of two acute angles is $5 : 4$, and the complement of one of these two angles is twice as large as the complement of the other. What is the sum of the degree measures of the two angles?

(A) 75 **(B)** 90 **(C)** 135 **(D)** 150 **(E)** 270

(Hint 1, Hint 2)

8. In the figure, $\ell_1 \parallel \ell_2$ and P lies strictly between them. Find x.



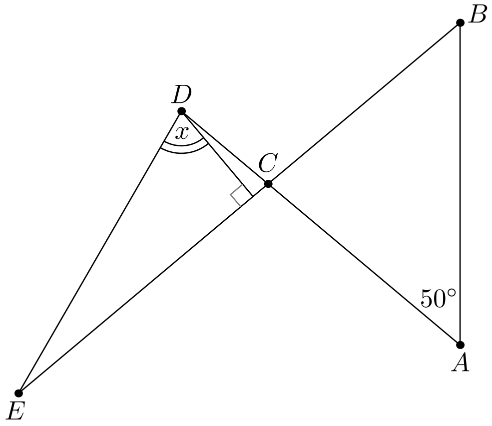
(Hint 1, Hint 2)

9. (2025 AMC 10A #6) In an equilateral triangle each interior angle is trisected by a pair of rays. The intersection of the interiors of the middle $20°$-angle at each vertex is the interior of a convex hexagon. What is the degree measure of the smallest angle of this hexagon?

(A) 80 **(B)** 90 **(C)** 100 **(D)** 110 **(E)** 120

(Hint 1, Hint 2)

10. Consider the figure below. Suppose $AC = BC$ and $DE = CE$. If $\angle CAB = 50°$, determine the value of the angle x.



(Hint 1, Hint 2)

Hints: Points, Lines, and Angles

Hint 1

1. Using opposite angles at the lower crossing, find b first.

   Now how can you get to c?

2. Draw the line AB and place X and Y on it.

   Set $XA = XB = a$ and $YA = YX = b$. Now what is a in terms of b?

3. Use the fact that the horizontal lines are parallel to find angle x.

   Focusing just on the second and third horizontal lines (which are parallel), find angle y.

4. You could find the interior angle at C first and then subtract from $180°$, but there is a one-step route.

   Which two angles of the triangle are the *remote* interior angles for the exterior angle x?

5. Do not try to find x, y, z separately. You cannot: the triangle is not pinned down.

   Each of x, y, z is an exterior angle of $\triangle ABC$. Write each one in terms of the interior angles.

6. Going through $(n-2) \cdot 180^\circ$ works, but the exterior angle is faster.

   At every vertex, interior plus exterior is 180°. What is one exterior angle here?

7. Since the ratio of the two acute angles is $5 : 4$, write the angles as $5x$ and $4x$.

   Their complements are

   $$90 - 5x$$

   and

   $$90 - 4x$$

   Which complement is larger?

8. P sits between the two parallel lines but on neither, so no rule reaches it directly.

   Draw a third line through P parallel to both. What do the $96°$ split into?

9. Each angle of the equilateral triangle is 60°, so trisecting gives angles of 20°.

   The sides of the hexagon lie on these trisector rays. Focus on one corner of the hexagon formed by two trisectors from two neighboring vertices of the big triangle. Can you make a triangle whose third side is one side of the original equilateral triangle?

10. Two isosceles triangles are hiding here. Start with $AC = BC$: what does that force about $\angle CAB$ and $\angle CBA$, and therefore about $\angle ACB$?

    Then notice D lies on line CA through C, and E on line CB through C. How must $\angle DCE$ relate to $\angle ACB$?

Hint 2

1. Opposite angles give b immediately.

   For c and a, use the fact that the horizontal lines are parallel: the crossing at the top line is a copy of the crossing at the bottom line.

2. Because $b = YA = YX = \frac{AX}{2} = \frac{a}{2}$ we have $a = 2b$. This means $XA = XB = 2b$.

   Now what is BY in terms of b?

   What is AB in terms of b?

3. We get that $x = 50^\circ$ and $y = 60^\circ$.

   Use the fact that angles x, y, z make a full angle of 360°.

4. The exterior angle theorem gives $x = 42° + 71°$ directly.

   Sanity check it the long way: the interior angle at C is $180° - 42° - 71°$, and x is $180°$ minus that. Do the two routes agree?

5. By the exterior angle theorem,

$$x = \angle B + \angle C, \qquad y = \angle C + \angle A, \qquad z = \angle A + \angle B.$$

   Add all three. Every interior angle appears exactly twice.

   What does that make the total, given the angle sum of a triangle?

6. Interior 156° means each exterior angle is $180^\circ - 156^\circ = 24^\circ$.

   The exterior angles of any convex polygon sum to 360°, and here they are all equal. So how many are there?

7. Since $5x$ is the larger angle, its complement is the smaller complement.

   So the complement of $4x$ is twice the complement of $5x$:

$$90 - 4x = 2(90 - 5x)$$

   Solve for x, then find the sum

$$5x + 4x$$

8. The auxiliary parallel through P splits $\angle APB$ into a piece equal to the angle at A and a piece equal to the angle at B. So

$$x + 2x = 96^\circ.$$

Solve for x.

9. Look at a corner of the hexagon formed by two neighboring trisectors. The triangle made with one side of the original equilateral triangle has two angles of 40°.

   So that hexagon angle is
   $$180^\circ - 40^\circ - 40^\circ$$

   Another type of hexagon angle comes from a triangle with two angles of 20°. What angle does that give, and which type is smaller?

10. Since $AC = BC$, we get $\angle CAB = \angle CBA = 50^\circ$, so $\angle ACB = 80^\circ$. Because CD and CE are the opposite rays of CA and CB, the vertical angle gives $\angle DCE = 80^\circ$.

    Now $DE = CE$ makes triangle CDE isosceles with its apex at E; the base angles at C and D are equal, and you know the one at C is 80°. Find the apex angle $\angle DEC$. Then, since $\overline{DK} \perp \overline{CE}$ makes $\triangle DKE$ right-angled at K, the angle x and $\angle DEC$ are complementary.

    So what is x?

Triangle Inequality, Pythagoras, and the Special Right Triangles

The triangle inequality

Not every three lengths can be the sides of a triangle.

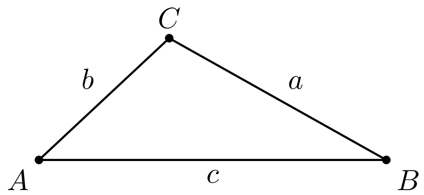
Main idea/Fact (Triangle inequality)

In any triangle with side lengths a, b, c, each side is shorter than the sum of the other two:

$$a < b + c, \qquad b < a + c, \qquad c < a + b.$$

Equivalently, if two sides are a and b, the third side c satisfies

$$|a - b| < c < a + b.$$



Why this is true. The straight path from A to B is the shortest one. Walking from A to B through the detour C covers $b + a$, so the direct distance c must be smaller: $c < a + b$. The same argument at the other two vertices gives the other two inequalities, and rearranging gives the two-sided form.

Main idea/Fact (Which inequality to check)

Only one of the three inequalities can fail: the one whose left side is the *longest* side. So in practice, identify the longest side and check it against the sum of the other two. If that passes, all three pass.

Example 7

Two sides of a triangle have lengths 4 and 9. How many integer values are possible for the third side?

Solution. Call the third side s. The two-sided form gives

$$9 - 4 < s < 9 + 4, \qquad \text{that is,} \qquad 5 < s < 13.$$

The integers strictly between 5 and 13 are

$$6, 7, 8, 9, 10, 11, 12,$$

so there are $\boxed{7}$ possible values.

Pitfall

The inequalities are strict. If $c = a + b$ exactly, the "triangle" collapses flat onto a line and has zero area. AMC problems often say "a triangle with positive area" precisely to rule this out, so do not count the endpoints.

Example 8 (AMC 10B 2006/10)

In a triangle with integer side lengths, one side is three times as long as a second side, and the length of the third side is 15. What is the greatest possible perimeter of the triangle?

Solution. Let the first side be x, so the sides are x, $3x$, and 15. To maximize the perimeter we want x as large as possible, and the binding constraint is the one on the longest side. If $3x$ is the longest side, the triangle inequality demands

$$3x < x + 15 \qquad \Longrightarrow \qquad x < \tfrac{15}{2}.$$

The greatest integer satisfying this is $x = 7$, and the perimeter is

$$7 + 21 + 15 = \boxed{43}.$$

Exercise 9. A triangle has sides of length 6.5 and 10, and its third side has integer length s. What is the smallest possible value of s?[2]

The Pythagorean theorem

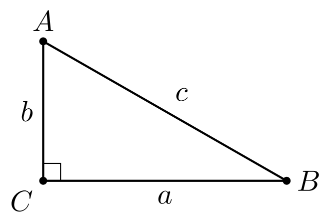
Definition (Legs and hypotenuse)

In a right triangle, the two sides forming the right angle are the **legs**, and the side opposite the right angle is the **hypotenuse**. The hypotenuse is always the longest side.

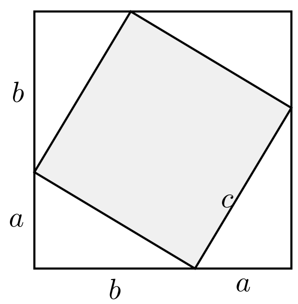
Main idea/Fact (Pythagorean theorem)

In a right triangle with legs a, b and hypotenuse c,

$$a^2 + b^2 = c^2.$$



Why this is true. Take a square of side $a + b$ and place four copies of the right triangle inside it, one in each corner, hypotenuses facing inward.



[2] We need $s > 10 - 6.5 = 3.5$, so the smallest integer is $s = 4$. Check the other direction: $4 < 16.5$, fine.

The four hypotenuses form the shaded inner quadrilateral. Its sides all have length c, and its angles are right angles: at each inner corner, the two acute angles of the triangle sum to $90°$, leaving exactly $90°$ for the shaded angle. So the shaded region is a square of area c^2. Now count the big square's area two ways:

$$(a+b)^2 = 4 \cdot \frac{1}{2}ab + c^2.$$

Expanding the left side gives $a^2 + 2ab + b^2 = 2ab + c^2$, and subtracting $2ab$ leaves

$$a^2 + b^2 = c^2.$$

Main idea/Fact (The converse)

The theorem runs backwards too: if a triangle's sides satisfy $a^2 + b^2 = c^2$, then the angle opposite c is a right angle. This is how the AMC hides right angles: a triangle with sides 3, 4, 5 is right whether or not the problem says so.

Main idea/Fact (Pythagorean triples worth memorizing)

Integer solutions of $a^2 + b^2 = c^2$ are called **Pythagorean triples**. Know these on sight:

$$(3,4,5), \quad (5,12,13), \quad (8,15,17), \quad (7,24,25), \quad (9,40,41), \quad (20,21,29).$$

Scaling a triple by any factor gives another: from $(3,4,5)$ come $(6,8,10)$, $(9,12,15)$, $(12,16,20)$, $(15,20,25)$, and from $(5,12,13)$ comes $(10,24,26)$. When two numbers of a triple appear in a problem, the third is being handed to you.

Pitfall

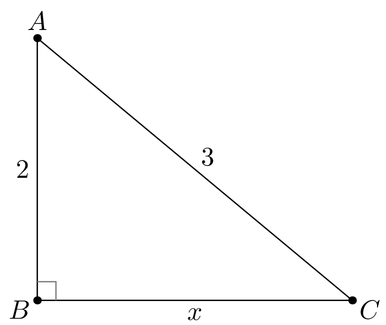
The largest number of the triple is always the hypotenuse. A triangle with legs 3 and 5 does not have hypotenuse 4; its hypotenuse is $\sqrt{34}$. And the theorem applies to right triangles only: for other triangles, $a^2 + b^2 \neq c^2$, and that failure is exactly what the converse detects.

Example 10

In right triangle ABC with $\angle B = 90°$, $AB = 2$ and $AC = 3$. Find BC.

Solution. The right angle is at B, so AC is the hypotenuse. Writing $BC = x$,

$$2^2 + x^2 = 3^2 \qquad \Longrightarrow \qquad x = \sqrt{9-4} = \boxed{\sqrt{5}}.$$



Example 11

A rectangle has sides 5 and 12. Find the length of its diagonal.

Solution. The diagonal is the hypotenuse of a right triangle with legs 5 and 12. Recognize the triple $(5, 12, 13)$:

$$\sqrt{5^2 + 12^2} = \sqrt{169} = \boxed{13}.$$

The 45-45-90 triangle

Definition (45-45-90)

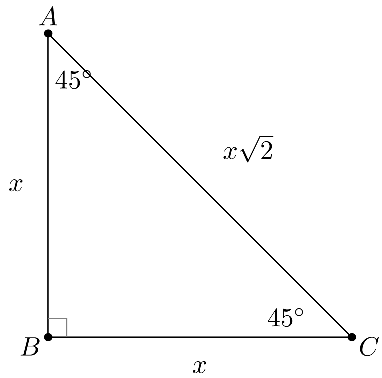
A **45-45-90** triangle is a right triangle whose other two angles are both 45°.

Main idea/Fact (Sides of a 45-45-90)

A 45-45-90 triangle is an isosceles right triangle: the two legs are equal. If each leg has length x, the hypotenuse is $x\sqrt{2}$, so the sides are in ratio

$$\underbrace{x}_{\text{leg}} : \underbrace{x}_{\text{leg}} : \underbrace{x\sqrt{2}}_{\text{hypotenuse}}.$$

Why this is true. The two base angles are both 45°. By the isosceles fact from Part I, equal angles force the opposite sides to be equal, so both legs have the same length x.



The right angle sits between the two legs, so Pythagoras gives the hypotenuse:

$$\sqrt{x^2 + x^2} = \sqrt{2x^2} = x\sqrt{2}.$$

Example 12

A 45-45-90 triangle has legs of length 7. How long is the hypotenuse?

Solution. Each leg is $x = 7$, so the hypotenuse is $x\sqrt{2} = \boxed{7\sqrt{2}}$.

Remark. The single most common appearance of this triangle is the *diagonal of a square.* A square of side s is two 45-45-90 triangles glued along the diagonal, so the diagonal is $s\sqrt{2}$. Read it backwards too: a square with diagonal d has side $d/\sqrt{2}$.

The 30-60-90 triangle

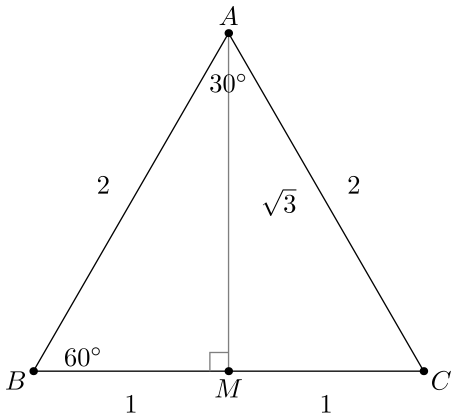
Definition (30-60-90)
A **30-60-90** triangle is a right triangle whose other two angles are $30°$ and $60°$.

Main idea/Fact (Sides of a 30-60-90)
In a 30-60-90 triangle, the side opposite the $30°$ angle is the shortest. If that side has length x, then

$$\underbrace{x}_{\text{opp }30°} : \underbrace{x\sqrt{3}}_{\text{opp }60°} : \underbrace{2x}_{\text{opp }90°}.$$

In words: the hypotenuse is *twice* the shortest side, and the remaining side is the shortest side times $\sqrt{3}$.

Why this is true. Take an equilateral triangle ABC with side length 2 and drop the height AM from A to BC.



Since ABC is equilateral, it is in particular isosceles with $AB = AC$, so by the Part I fact the height AM is also the median and the angle bisector from A. Therefore M is the midpoint of BC, giving $BM = 1$, and the apex angle is split into two halves of $30°$ each.

Now look at triangle ABM. Its angles are $90°$ at M, $60°$ at B, and $30°$ at A, so it is a 30-60-90 triangle. Its sides are

$$BM = 1 \text{ (opposite } 30°), \qquad AB = 2 \text{ (opposite } 90°),$$

and by the Pythagorean theorem the last side is

$$AM = \sqrt{AB^2 - BM^2} = \sqrt{2^2 - 1^2} = \sqrt{3} \text{ (opposite } 60°).$$

So the sides are in ratio $1 : \sqrt{3} : 2$. Scaling every length by the same factor x keeps the angles the same, so a general 30-60-90 triangle has sides $x : x\sqrt{3} : 2x$.

Example 13
In a 30-60-90 triangle, the side opposite the $30°$ angle has length 5. Find the other two sides.

Solution. The shortest side is $x = 5$, so the side opposite $60°$ is $x\sqrt{3} = \boxed{5\sqrt{3}}$ and the hypotenuse is $2x = \boxed{10}$.

Main idea/Fact (The two special right triangles)

Both ratios are worth memorizing, and both are read *across from the angle*:

$$\textbf{30-60-90:}\quad 1 : \sqrt{3} : 2, \qquad\qquad \textbf{45-45-90:}\quad 1 : 1 : \sqrt{2}.$$

The hypotenuse (across from the $90°$) is always the largest number.

Pitfall

In a 30-60-90 triangle, the hypotenuse is *twice the shortest side*, not $\sqrt{3}$ times it. The $\sqrt{3}$ belongs to the *middle* side (opposite $60°$). Mixing these up is the most common mistake in the whole topic:

$$\text{hypotenuse} = 2x \quad \text{(correct)}, \qquad \text{hypotenuse} = x\sqrt{3} \quad \text{(wrong)}.$$

Main idea/Fact (Where these two triangles come from)

Nearly every appearance on the AMC is one of these:

- Cut a square along a diagonal, or cut it with a $45°$ line: 45-45-90.
- Drop the altitude in an equilateral triangle, or halve one: 30-60-90.
- A regular hexagon cut into six pieces from the center: six equilateral triangles, so 30-60-90 again.
- Any right triangle whose given angle is $30°$, $45°$, or $60°$.

Spotting the source is usually the whole problem.

Summary of Part II

Main idea/Fact (Lengths in a triangle)

- Triangle inequality: each side is strictly less than the sum of the other two, so a third side satisfies $|a - b| < c < a + b$. Only the longest side's inequality can fail, so check that one.
- Strict means strict. A degenerate "triangle" has zero area, and the AMC rules it out with the phrase "positive area".
- $a^2 + b^2 = c^2$ in right triangles, in both directions. The converse is how hidden right angles are detected.
- Know $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(9, 40, 41)$, $(20, 21, 29)$ and their multiples on sight.
- 45-45-90 has ratio $1 : 1 : \sqrt{2}$; 30-60-90 has ratio $1 : \sqrt{3} : 2$. Read across from the angle, and remember the hypotenuse of a 30-60-90 is $2x$, never $x\sqrt{3}$.

Problems: Triangle Inequality, Pythagoras, Special Right Triangles

1. The sides of a triangle are 6, 11, and x, where x is a positive integer. How many possible values of x are there?

(Hint 1, Hint 2)

2. (2000 AMC 10 #10) The sides of a triangle with positive area have lengths 4, 6, and x. The sides of a second triangle with positive area have lengths 4, 6, and y. What is the smallest positive number that is *not* a possible value of $|x-y|$?

(A) 2 **(B)** 3 **(C)** 4 **(D)** 6 **(E)** 8

(Hint 1, Hint 2)

3. (2017 AMC 10A #10) Joy has 30 thin rods, one each of every integer length from 1 cm through 30 cm. She places the rods with lengths 3 cm, 7 cm, and 15 cm on a table. She then wants to choose a fourth rod that she can put with these three to form a quadrilateral with positive area. How many of the remaining rods can she choose as the fourth rod?

(A) 16 **(B)** 17 **(C)** 18 **(D)** 19 **(E)** 20

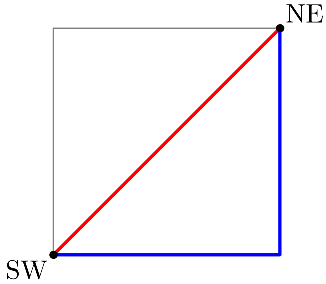
(Hint 1, Hint 2)

4. (2009 AMC 10B #10) A flagpole is originally 5 meters tall. A hurricane snaps the flagpole at a point x meters above the ground so that the upper part, still attached to the stump, touches the ground 1 meter away from the base. What is x?

(A) 2.0 **(B)** 2.1 **(C)** 2.2 **(D)** 2.3 **(E)** 2.4

(Hint 1, Hint 2)

5. (2017 AMC 10A #7) Jerry and Silvia wanted to go from the southwest corner of a square field to the northeast corner. Jerry walked due east and then due north to reach the goal, but Silvia headed northeast and reached the goal walking in a straight line. Which of the following is closest to how much shorter Silvia's trip was, compared to Jerry's trip?



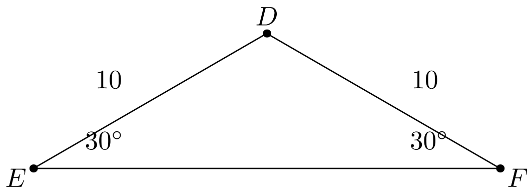
(A) 30% **(B)** 40% **(C)** 50% **(D)** 60% **(E)** 70%

(Hint 1, Hint 2)

6. A ladder of length 10 leans against a vertical wall and makes a 60° angle with the level ground. How high up the wall does the top of the ladder reach?

(Hint 1, Hint 2)

7. In $\triangle DEF$ we have $DE = DF = 10$ and $\angle DEF = \angle DFE = 30°$. Find the perimeter of $\triangle DEF$.



(Hint 1, Hint 2)

8. An equilateral triangle has altitude 9. What is its side length?

(Hint 1, Hint 2)

Hints: Triangle Inequality, Pythagoras, Special Right Triangles

Hint 1

1. By the triangle inequality, the third side must be smaller than the sum of the other two and larger than their difference.

   With sides 6 and 11, strictly between which two numbers must x lie?

2. Each unknown side must satisfy the triangle inequality with 4 and 6, and y obeys the exact same bound as x.

   Strictly between which two values must x (and y) lie?

3. Closing four sticks into a quadrilateral with real area is the triangle inequality with one extra side: each side must be shorter than the sum of the other three. Let the new rod have length d.

   Which length is in danger of being ”too long” to close up, only d, or also the existing 15?

4. Picture the broken flagpole as a right triangle: the standing stump (height x) is one leg, the 1 meter along the ground is the other leg, and the leaning top piece is the hypotenuse. The whole pole was 5 meters long.

   In terms of x, how long is that leaning top piece?

5. When a comparison doesn't depend on the actual size, you may pick a convenient one. Let the square's side be 1. Then Jerry walks two sides, a length of $1 + 1$, while Silvia walks the diagonal, the hypotenuse of a 45-45-90 triangle with legs 1.

   What is the length of that diagonal?

6. The ladder, the wall, and the ground form a right triangle, right-angled where the wall meets the ground. One acute angle is $60°$.

   What are all three angles, and which side is the ladder?

7. The triangle is isosceles with apex D. Drop the altitude from D to EF; in an isosceles triangle that altitude is also the median, so it cuts EF exactly in half.

   What are the three angles of each half, and which side of that half do you already know?

8. The altitude of an equilateral triangle splits it into two 30-60-90 triangles.

   In the ratio $1 : \sqrt{3} : 2$, which entry is the altitude and which is the full side?

Hint 2

1. The inequality gives $11 - 6 < x < 11 + 6$, that is $5 < x < 17$. The inequalities are strict, so 5 and 17 themselves are not allowed.

   How many integers lie strictly between 5 and 17?

2. The triangle inequality forces $6 - 4 < x < 6 + 4$, i.e. $2 < x < 10$, and the same for y. The largest $|x - y|$ could approach is the full width of that interval, but the endpoints are excluded, so it can never quite reach that width.

   What is the smallest positive value that $|x - y|$ can therefore never equal?

3. Both lengths can be too long: you need $d < 3 + 7 + 15$ and $15 < 3 + 7 + d$. Together these force $5 < d < 25$, so d ranges over the integers 6 through 24.

   Count those integers, then recall that two of them (7 and 15) are already on the table. How many usable rods remain?

4. The leaning piece is $5 - x$, so the Pythagorean theorem says

$$x^2 + 1^2 = (5 - x)^2.$$

Expand the right-hand side and watch the x^2 cancel from both sides. What linear equation in x is left, and what value of x does it give?

5. The diagonal is $\sqrt{2} \approx 1.41$, so Silvia saves $2 - \sqrt{2}$ of walking. "How much shorter" as a fraction of Jerry's trip is
$$\frac{2-\sqrt{2}}{2}.$$
Work out that decimal. Which answer choice is it closest to?

6. The angles are 30°, 60°, 90°, and the ladder is the hypotenuse, so $2x = 10$ and the shortest side is $x = 5$. That shortest side is opposite the 30°, which is the distance along the ground.

   The wall height is opposite the 60°. What is $x\sqrt{3}$?

7. Each half is a 30-60-90 triangle with hypotenuse 10, so the shortest side (opposite 30°, which is the altitude) is 5, and the side opposite 60° is $5\sqrt{3}$. That last one is *half* of EF.

   So what is EF, and what is $10 + 10 + EF$?

8. Write the side as $2x$. Then half the side is x (opposite $30°$) and the altitude is $x\sqrt{3}$ (opposite $60°$).

   Solve $x\sqrt{3} = 9$, rationalize, and double the answer to get the side.

Area, Perimeter, and Known Figures

Perimeter

Definition (Perimeter)

The **perimeter** of a polygon is the sum of all its side lengths. For a triangle with sides a, b, c, the perimeter is $a + b + c$.

Half the perimeter is the **semi-perimeter**, written s. For a triangle, $s = \frac{a+b+c}{2}$. It looks like a strange thing to name, but several later formulas are cleanest in terms of s, so we introduce the notation now.

Main idea/Fact (Perimeter of a path, not of a region)

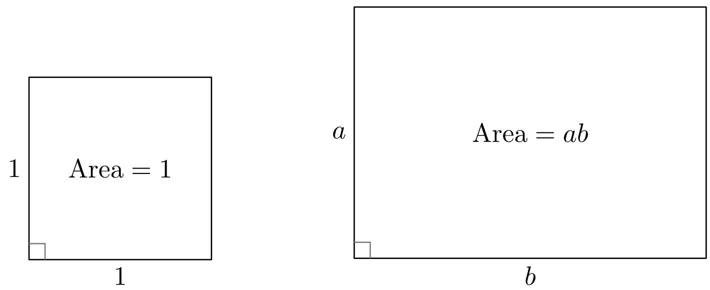
Perimeter counts the boundary you would walk. For a region cut out of a larger figure, every straight cut you make adds new boundary. This is why the perimeter of a half-disk is not half the circumference, as we will see below.

Area

Definition (Area)

Area measures how much of the plane a region covers. The unit of measurement is the unit square: a square whose sides have length 1 has area 1. We write $[X]$ for the area of a figure X.

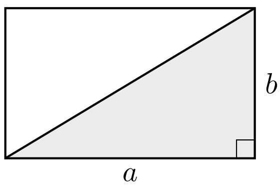
From this single convention, every formula follows. A square of side length a can be cut into an $a \times a$ grid of unit squares, so its area is a^2. A rectangle with side lengths a and b can be cut into an $a \times b$ grid, so its area is ab.



Main idea/Fact (Square and rectangle)

A square of side a has area a^2 and perimeter $4a$. A rectangle with adjacent sides a and b has area ab and perimeter $2(a + b)$.

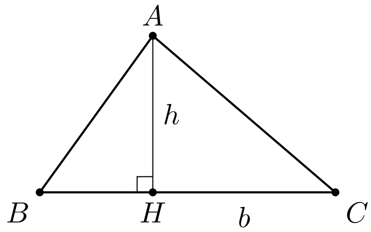
But most shapes do not have perpendicular sides, and the whole art of area is reducing them to shapes that do. The first reduction is the right triangle.



The diagonal cuts the rectangle into two identical right triangles, so a right triangle with legs a and b has area $\frac{1}{2}ab$. Now we can handle every triangle.

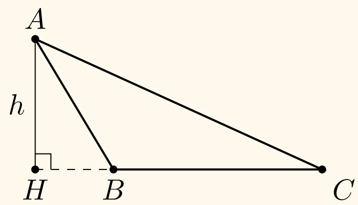
> **Main idea/Fact** (Area of a triangle)
>
> A triangle with base b and height h has area
>
> $$[\triangle] = \frac{1}{2}bh,$$
>
> where the **height** (or altitude) is the perpendicular distance from the opposite vertex to the line of the base.

Why this is true. Drop the perpendicular from A to the base BC, meeting it at H.



This splits $\triangle ABC$ into two right triangles, and each is half of a rectangle:

$$[ABC] = [ABH] + [AHC] = \frac{1}{2}\,BH \cdot h + \frac{1}{2}\,HC \cdot h = \frac{1}{2}(BH + HC)\,h = \frac{1}{2}bh.$$

> **Pitfall**
>
> If the triangle is obtuse, the foot H can land *outside* the base:
>
> 
>
>
> The formula still holds (now $[ABC] = [AHC] - [AHB]$, a subtraction instead of a sum), but the height is still the *perpendicular* distance to the line of the base.

Any side can be the base

> **Main idea/Fact** (Three base-height pairs, one area)
>
> A triangle has three base-height pairs, and all three give the same area:
>
> $$\frac{1}{2}\,a\,h_a = \frac{1}{2}\,b\,h_b = \frac{1}{2}\,c\,h_c.$$
>
> Computing the area with one pair and setting it equal to another pair is the standard AMC move for finding an unknown height or an unknown side.

Example 14

A right triangle has legs 6 and 8. Find the height to the hypotenuse.

Solution. First the hypotenuse: $(6, 8, 10)$ is twice $(3, 4, 5)$, so it is 10.

Now compute the area two ways. Using the legs, which are perpendicular to each other and therefore already a base-height pair,

$$[\triangle] = \frac{1}{2} \cdot 6 \cdot 8 = 24.$$

Using the hypotenuse as base with unknown height h,

$$[\triangle] = \frac{1}{2} \cdot 10 \cdot h = 5h.$$

The two expressions describe the same area, so $5h = 24$ and

$$h = \boxed{4.8}.$$

Remark. That solution used almost everything so far at once: a memorized triple, the area formula, and the same quantity counted twice. This "area two ways" move appears on the AMC every single year, and it is the fastest route to any altitude that is not one of the sides.

Equilateral and isosceles triangles

These two deserve their own treatment because their areas come up constantly and because both are computed the same way: drop the altitude from the apex and use Part II.

Main idea/Fact (Area of an equilateral triangle)

An equilateral triangle of side s has altitude $\dfrac{s\sqrt{3}}{2}$ and area

$$[\triangle] = \frac{s^2\sqrt{3}}{4}.$$

Why this is true. The altitude splits it into two 30-60-90 triangles with shortest side $s/2$ (opposite the $30°$) and hypotenuse s. The altitude is opposite the $60°$, so it equals $\frac{s}{2}\sqrt{3}$. Then

$$[\triangle] = \frac{1}{2} \cdot s \cdot \frac{s\sqrt{3}}{2} = \frac{s^2\sqrt{3}}{4}.$$

Main idea/Fact (Area of an isosceles triangle)

For an isosceles triangle with equal sides ℓ and base b, drop the altitude from the apex. It bisects the base, so Pythagoras gives

$$h = \sqrt{\ell^2 - \left(\frac{b}{2}\right)^2}, \qquad \text{and} \qquad [\triangle] = \frac{1}{2}\, b \sqrt{\ell^2 - \frac{b^2}{4}}.$$

Do not memorize the second formula. Memorize the *move*: drop the apex altitude, use that it hits the midpoint, then Pythagoras.

Example 15

An isosceles triangle has equal sides of length 25 and base 14. Find its area.

Solution. The apex altitude bisects the base into two pieces of 7. The right triangle formed has hypotenuse 25 and one leg 7, and $(7, 24, 25)$ is on our list, so $h = 24$. Then

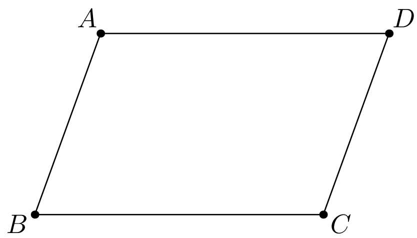
$$[\triangle] = \frac{1}{2} \cdot 14 \cdot 24 = \boxed{168}.$$

Quadrilaterals

We now meet the standard four-sided shapes, and we compute every area by cutting into triangles and rectangles.

Definition (Parallelogram)

A quadrilateral whose opposite sides are parallel.



Main idea/Fact (Parallelogram properties)

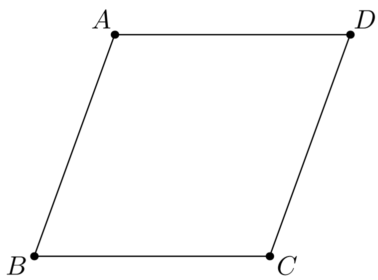
In a parallelogram, opposite sides are equal, opposite angles are equal, adjacent angles sum to $180°$ (they are co-interior angles between parallel lines), and the diagonals cut each other in half. The proofs use congruent triangles, which is the Week 2 Geometry lecture.

Definition (Rectangle and square)

A **rectangle** is a parallelogram with four right angles. A **square** is a rectangle with all four sides equal.

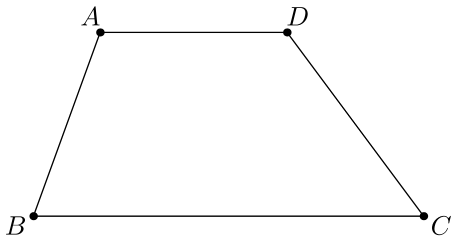
Definition (Rhombus)

A quadrilateral with all four sides equal. A rhombus is automatically a parallelogram.



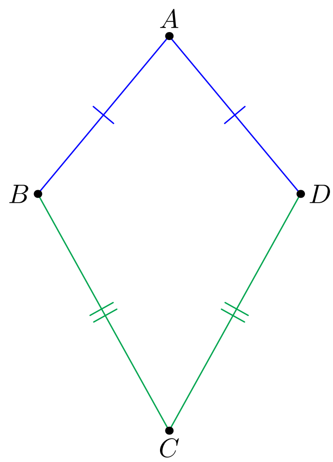
Definition (Trapezoid)

A quadrilateral with exactly one pair of parallel sides. The parallel sides are the **bases**, and the perpendicular distance between them is the **height**.



Definition (Kite)

A quadrilateral with two pairs of equal adjacent sides. The line through the two vertices where unequal sides meet is an axis of symmetry, and it follows that the diagonals of a kite are perpendicular.



Main idea/Fact (Areas of quadrilaterals)

With b a base, h the height on it, b_1, b_2 the parallel sides of a trapezoid, and d_1, d_2 diagonals:

$$\text{square: } a^2 \qquad \text{rectangle: } ab \qquad \text{parallelogram: } bh$$
$$\text{trapezoid: } \frac{b_1+b_2}{2}\cdot h \qquad \text{kite and rhombus: } \frac{1}{2}d_1d_2$$

A rhombus is both a parallelogram and a kite, so both bh and $\frac{1}{2}d_1d_2$ apply to it. Choose whichever the problem hands you.

Why these are true. Each formula is a triangle cut.

- **Parallelogram.** A diagonal cuts it into two triangles of equal area, each with base b and height h, so the total is $2 \cdot \frac{1}{2}bh = bh$.
- **Trapezoid.** A diagonal cuts it into two triangles, one with base b_1 and one with base b_2, both with height h. The total is $\frac{1}{2}b_1h + \frac{1}{2}b_2h = \frac{b_1+b_2}{2} \cdot h$. Note that $\frac{b_1+b_2}{2}$ is the

length of the **midline**, the segment joining the midpoints of the non-parallel sides, so the area is also midline times height.

- **Kite.** The axis diagonal d_1 cuts the kite into two triangles with base d_1. Since the diagonals are perpendicular, each triangle's height is a piece of d_2, and the two heights sum to d_2. The total area is $\frac{1}{2}d_1d_2$.

Pitfall

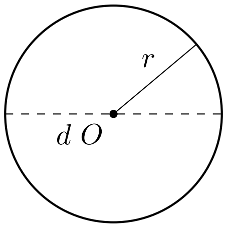
The height of a parallelogram or trapezoid is the *perpendicular* distance between the parallel sides. The slanted side is longer than the height, and using it is a classic way to overcount an area.

Exercise 16. A trapezoid has parallel sides 7 and 13 and height 4. Find its area.[3]

Circles

Definition (Circle)

A **circle** is the set of all points at a fixed distance, the **radius** r, from a fixed center. A **diameter** is a segment through the center with both endpoints on the circle; its length is $d = 2r$. The distance once around the circle is its **circumference**.



Main idea/Fact (Circumference and area)

For a circle of radius r,

$$\text{circumference} = 2\pi r = \pi d, \qquad \text{area} = \pi r^2,$$

where $\pi = 3.14159\ldots$ is the same constant in both formulas.

Usually you will write the answer in terms of π: you will write 16π, not 50.27. That means circle computations are exact and clean, and π usually just rides along or cancels.

The only times you need to remember roughly what π is equal to (knowing it is roughly 3.14159 will almost always suffice) is when you need to *estimate* the area, perimeter, or some value that has π in it.

Pitfall

The perimeter of a half-disk (a semicircular region) is *not* half the circumference. The boundary consists of the curved half, πr, plus the straight diameter, $2r$. Forgetting the diameter is one of the most common Week 0 mistakes.

[3] $\frac{7+13}{2} \cdot 4 = 10 \cdot 4 = 40$.

Exercise 17. Find the perimeter and the area of a half-disk of radius 4.[4]

Sanity checks

Main idea/Fact (Before you bubble)

- Any triangle you produce must have angles summing to 180° and sides obeying the triangle inequality. Check both.
- The hypotenuse must be the longest side of its triangle, and any height must be shorter than the slanted sides next to it.
- Perimeters are in plain units, areas in square units. If a length doubles, the perimeter doubles but the area quadruples.
- The figure is not to scale. If your answer contradicts the picture but follows from the givens, trust the givens.

Summary of Part III

Main idea/Fact (Area and perimeter)

- Perimeter is the boundary you walk; cutting a region adds new boundary.
- Areas come from cutting into rectangles and triangles: triangle $\frac{1}{2}bh$, parallelogram bh, trapezoid $\frac{b_1+b_2}{2}h$, kite and rhombus $\frac{1}{2}d_1d_2$, circle πr^2 with circumference $2\pi r$.
- Equilateral of side s: altitude $\frac{s\sqrt{3}}{2}$, area $\frac{s^2\sqrt{3}}{4}$. Isosceles: drop the apex altitude, use the midpoint, then Pythagoras.
- The height is always the perpendicular distance, and any side of a triangle can serve as the base. Counting one area two ways solves for unknown heights and unknown sides.

[4] Perimeter: $\pi \cdot 4 + 2 \cdot 4 = 4\pi + 8$. Area: $\frac{1}{2}\pi \cdot 4^2 = 8\pi$.

Problems: Area, Perimeter, and Known Figures

1. A square and an equilateral triangle each have perimeter 36. Which one has the larger area, and by how much?

(Hint 1, Hint 2)

2. An equilateral triangle has side length 6. What is its area?

(Hint 1, Hint 2)

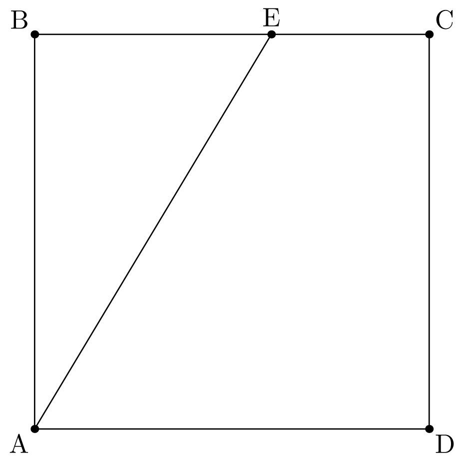
3. An isosceles triangle has two equal sides of length 13 and a base of length 10. What is its area?

(Hint 1, Hint 2)

4. In $\triangle ABC$ we have $AB = 15$, and the altitude from C to $\overline{AB}$ has length 8. The altitude from A to $\overline{BC}$ has length 10. Find BC.

(Hint 1, Hint 2)

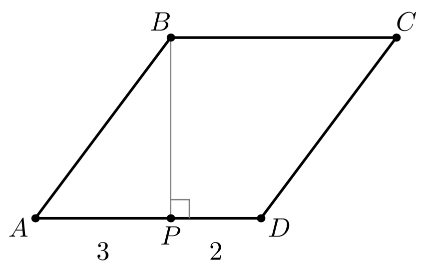
5. (2013 AMC 10A #3) Square $ABCD$ has side length 10. Point E is on $\overline{BC}$, and the area of $\triangle ABE$ is 40. What is BE?



(A) 4 **(B)** 5 **(C)** 6 **(D)** 7 **(E)** 8

(Hint 1, Hint 2)

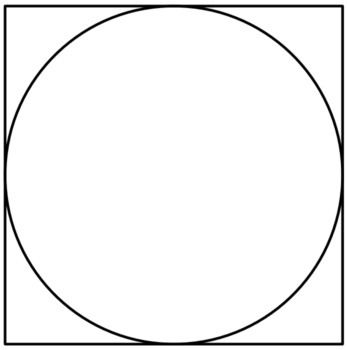
6. (2022 AMC 10B #2) In rhombus $ABCD$, point P lies on segment $\overline{AD}$ so that $\overline{BP} \perp \overline{AD}$, $AP = 3$, and $PD = 2$. What is the area of $ABCD$?



(A) $3\sqrt{5}$ **(B)** 10 **(C)** $6\sqrt{5}$ **(D)** 20 **(E)** 25

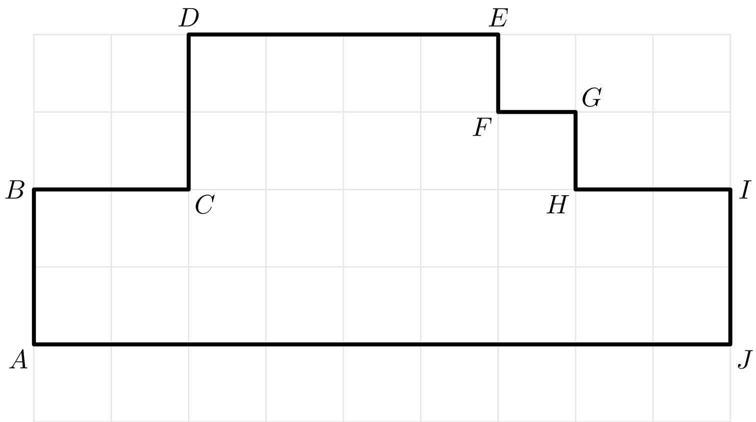
(Hint 1, Hint 2)

7. A circle is inscribed in a square of side length 10, touching all four sides. What is the area of the region inside the square but outside the circle?



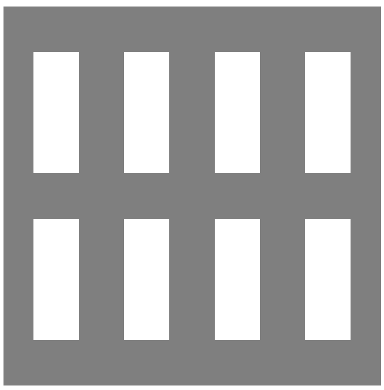
(Hint 1, Hint 2)

8. Each little square in the grid is a 1×1 square. Find the area and perimeter of the shape $ABCDEFGHIJ$:



(Hint 1, Hint 2)

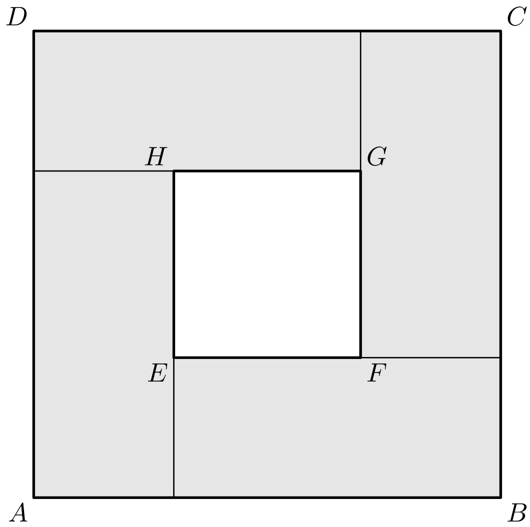
9. (2014 AMC 10B #5) Doug constructs a square window using 8 equal-size panes of glass, as shown. The ratio of the height to width for each pane is $5 : 2$, and the borders around and between the panes are 2 inches wide. In inches, what is the side length of the square window?



(A) 26 **(B)** 28 **(C)** 30 **(D)** 32 **(E)** 34

(Hint 1, Hint 2)

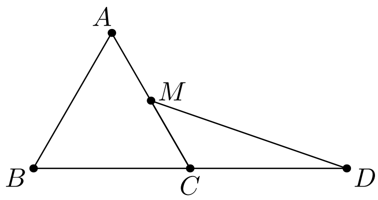
10. The square $ABCD$ has 4 identical rectangles inside it like in the picture below:



Each of the smaller rectangles has perimeter 20 and area 21. What is the area of $ABCD$? What about the inner square $EFGH$?

(Hint 1, Hint 2)

11. (2005 AMC 10B #14) Equilateral $\triangle ABC$ has side length 2, M is the midpoint of $\overline{AC}$, and C is the midpoint of $\overline{BD}$. What is the area of $\triangle CDM$?



(A) $\frac{\sqrt{2}}{2}$ **(B)** $\frac{3}{4}$ **(C)** $\frac{\sqrt{3}}{2}$ **(D)** 1 **(E)** $\sqrt{2}$

(Hint 1, Hint 2)

Hints: Area, Perimeter, and Known Figures

Hint 1

1. Perimeter first, then side length. A square has 4 equal sides and an equilateral triangle has 3.

   Once you have both side lengths, which area formula do you need for each?

2. Area needs a base and a perpendicular height. Take a side as the base, and drop the altitude to it; since the triangle is equilateral (hence isosceles), that altitude lands on the midpoint of the base.

   What kind of right triangle does the altitude create, and what are its three angles?

3. Area needs a base and a perpendicular height. Use the base of length 10 and drop the altitude from the apex. In an isosceles triangle that altitude is also the median.

   So where exactly does it meet the base, and what right triangle does it form?

4. You are given two different base-height pairs for the *same* triangle, and one of them is complete.

   Compute the area from the complete pair first.

5. Use BE as the base of $\triangle ABE$.

   Since BE lies on the side BC of the square, what is the height from A to the line containing BE?

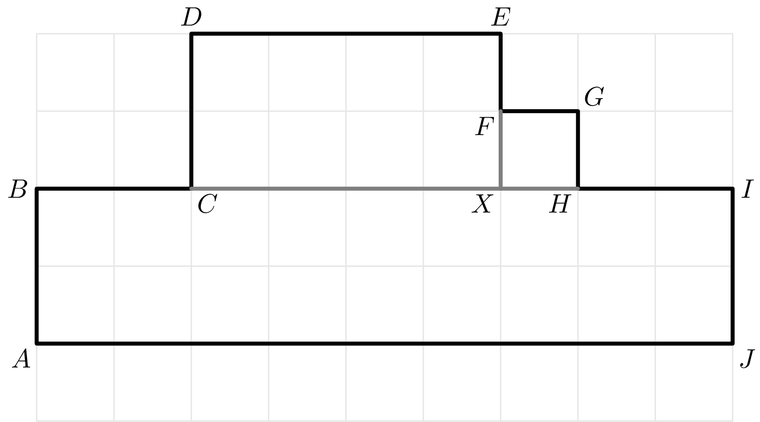
6. All four sides of a rhombus are equal, so $AB = AD = AP + PD = 5$. Now focus on triangle ABP: it has a right angle at P, a leg $AP = 3$, and hypotenuse $AB = 5$.

   Does that combination remind you of a famous right triangle?

7. The circle touches all four sides, so its diameter is exactly the side length of the square. What is the radius, and what are the two areas you need to subtract?

8. For the area, you can either break the shape up into rectangles or count how many squares there are.

   Let EF and HI meet at X. Calculate the areas of $ABIJ$, $XHGF$, $XCDE$.

   For the perimeter, the main idea is that the number of times you go "up" in the shape is the same number of times you go "down" in the shape. Same with "left" and "right".

9. The panes all have the same ratio, so write the width of each pane as $2x$ and the height of each pane as $5x$.

   Now count the total width and the total height of the whole window, including the 2-inch borders. What expression do you get for each?

10. Label the sidelengths of the smaller rectangles with x (for the larger side) and y (for the smaller side).

    The condition becomes $2x + 2y = 20$ and $xy = 21$.

    What is the sidelength of the square?

11. Aim for area $= \frac{1}{2} \cdot \text{base} \cdot \text{height}$ with CD as the base. Since C is the midpoint of $\overline{BD}$ and $BC = 2$, the base CD is forced. The height is the vertical drop from M down to line BD.

    How does M's height above BD compare to A's height above BD?

Hint 2

1. The square has side 9 and area 81. The equilateral triangle has side 12, so by the $\frac{s^2\sqrt{3}}{4}$ formula its area is $36\sqrt{3}$.

   Since $\sqrt{3} \approx 1.732$, estimate $36\sqrt{3}$ and compare to 81. What is the exact difference?

2. The altitude splits the base into two halves of 3, making a 30-60-90 triangle whose short leg (opposite the $30°$) is 3. Using the $1 : \sqrt{3} : 2$ ratio, the altitude (opposite the $60°$) is $3\sqrt{3}$.

   With base 6 and this height, what does $\frac{1}{2} \cdot 6 \cdot 3\sqrt{3}$ come out to?

3. The altitude meets the midpoint, splitting the base into two segments of 5 and forming a right triangle with hypotenuse 13 and one leg 5. The Pythagorean theorem gives the altitude as $\sqrt{13^2 - 5^2} = 12$.

   With base 10 and height 12, what is $\frac{1}{2} \cdot 10 \cdot 12$?

4. Using AB as the base, $[ABC] = \frac{1}{2} \cdot 15 \cdot 8 = 60$.

   Now use BC as the base with height 10: $\frac{1}{2} \cdot BC \cdot 10 = 60$. Solve for BC.

5. The height of $\triangle ABE$ is 10, because the square has side length 10.

   So the area condition gives

$$\frac{1}{2} \cdot BE \cdot 10 = 40$$

   Now solve for BE.

6. It is a 3-4-5 triangle, so $BP = \sqrt{5^2 - 3^2} = 4$. The area of a rhombus is base times height; take AD as the base and BP as the height.

   What is $AD \cdot BP$?

7. The diameter is 10, so $r = 5$. The square has area $10^2 = 100$ and the circle has area $\pi \cdot 5^2 = 25\pi$.

   What is the difference, left in terms of π?

8. From A (lowest point) to D (highest point) you're going to go up 4 times and so you're also going down 4 times. From A (leftmost point) to J (rightmost point) you'll go right 9 times.

   Therefore the perimeter is $4 \cdot 2 + 9 \cdot 2$.

9. There are 4 panes across, so the total width is

$$4(2x) + 5 \cdot 2 = 8x + 10$$

There are 2 panes vertically, so the total height is

$$2(5x) + 3 \cdot 2 = 10x + 6$$

Since the whole window is a square, these two expressions must be equal. What value of x makes

$$8x + 10 = 10x + 6$$

and then what is the side length?

10. The area of the square is $(x+y)^2 = 10^2 = 100$.

    Subtract the areas of the rectangles from the square to get the area of $EFGH$.

    The area of the inner square is $100 - 4 \cdot (\textit{Area of Rectangle})$.

11. The base is $CD = 2$ (since C is the midpoint of $\overline{BD}$). Because M is the midpoint of $\overline{AC}$, its height above line BD is exactly half of A's height. The equilateral triangle of side 2 has altitude $\sqrt{3}$.

    So how high above BD is M, and what does $\frac{1}{2} \cdot 2 \cdot$ (that height) give?

Shef Scholars

Ensuring no Ramanujan or Mirzakhani is ever missed again